

disk. The Xbox 360 would then read this mathematical template, send it to its processor where one of its six hardware threads would process and produce the required vertices of the tree, based on the mathematical template. These vertices would then be fed to the GPU for rendering and output—greatly reducing art content, developer costs, and even bandwidth requirement.

Such a procedural approach is not limited to geometry. It can also be used for animating game objects: specifically, the game developer need only define key frames for an animation, letting the Xenon generate the transition frames for completing the animation. The processor can also be fed key differences between two frames—for example, only certain parts of an object might change between two consecutive frames, the processor can be fed only the difference frames greatly reducing the content requirement and alleviating bandwidth needs to push the content.

Finally, the Xenon can tessellate an object: A 3D object within a game is represented via vertices and polygons. What might be a smooth soccer ball in real life is composed of a collection of triangles within a game—the smoother you want the ball to look, the greater the number of vertices needed. It takes more data to represent an object as a collection of triangles, than it does to represent it as a bunch of smooth curves. Tessellation is the process of breaking up a curve into a bunch of representative triangles. The Xenon can perform tessellation in hardware, thus significantly accelerating the process. A game can thus ship with curve information on an object, and take less space; the Xenon can subsequently use that information to generate the final object. This real-time tessellation can also be used for dynamically throttling the level of detail on an object: take the soccer ball as an example—when viewed from a distance, the number of polygons which comprise the ball can be reduced, as you approach the ball, the polygons can be automatically increased to give the ball a smoother look.

Currently, not a lot of titles use the procedural tools afforded by the Xenon's architecture. The RPG title *Oblivion* is perhaps the only one to use the trick to some effect. It employs a third party tool termed SpeedTree which procedurally generates the trees and the foliage which populate *Oblivion*'s vast landscape. But even *Oblivion* does not use it in real-time. Perhaps as time goes on, more titles will employ the idea.



Oblivion showcases procedurally generated trees and foliage

A Unified Approach

Microsoft decided to award ATI the contract for the Xbox 360's graphics processing unit (GPU). ATI is thus making the graphics part for two consoles this generation—the C1 or Xenos for Microsoft's Xbox 360 and the "Hollywood" chip for the Nintendo console, called Wii (pronounced "We").

Xenos, much like its neighbour Xenon, is a unique chip in many aspects.

On the PC, a graphics card divides its real-estate between vertex shaders and pixel shaders—hardware units which act on vertices of a 3D object and on individual pixels, respectively. This traditional architecture has its limitations. In a given game scene, it is never clear how much the vertex shader or the pixel shade would be employed, although current titles tend to lean heavily on the pixel parts. Thus, efficiency comes to question: a scenario in which a pixel pipeline is waiting for the vertex shaders to finish, or vice-versa is quite common. This leads to idle-time, wherein either of the two shader units is in essence twiddling its thumbs, waiting for the other to finish. This is a waste of power, not to mention of the transistors which comprise the idling unit.

A workaround is proposed via the 360's GPU in the form of unified shader architecture. Here, instead of specific shader units for each type of data, the GPU employs transistors which can handle both manner of tasks—pixel and vertex. Balancing and juggling a game's scene becomes much easier: for a game scene employing lots of special effects, the processor can put more shaders for pixel work and less on constructing the geometry of the scene, and so on.

A unified approach to shaders is very much on the cards for the PC space as well.

A shader is essentially a vector unit, much like the SIMD (pronounced "simdy") units of a PC processor. As such, shaders can carry out mathematically intense tasks not just restricted to graphics. The in-thing nowadays is physics acceleration on a chip. We currently have two distinct proposals for such a tactic—a dedicated physics processor sitting on a PCI bus, or employing a GPU to perform physics acceleration. Indeed, ATI has also proposed use of its GPU to handle audio tasks. All of this is also doable on the Xenos. Furthermore, the ATI part can also push and pull content to the shared 512 MB of memory. This allows the unit to work on data, push the result to memory and pull it back to another shader unit for additional processing and physics tasks. The GPU's unified approach to shaders, the multithreaded architecture of the 360's processor and the fact that both the CPU and the GPU share the 1 MB of L2 cache and the 512MB of system memory, all add to produce an extremely flexible architecture: the Xbox 360 can load balance—on its six hardware threads and its 48 shader units—tasks pertaining to procedural synthesis, pixel shading, vertex shading, physics, AI, and animation.

Such flexibility and interconnectedness keeps the entire system busy and increases its efficiency.

Apart from the unified shaders, the GPU also employs a very unique tool—embedded memory.